

Project Zenith: The Celestial Eyes

"Point anywhere on Earth and instantly understand everything happening above you."

Vision

Project Zenith: Celestial Lens is a real-time celestial intelligence platform designed to answer a simple question:

"What can I actually see in the sky right now, and what should I watch for next?"

The platform combines astronomical telemetry, satellite tracking, weather intelligence, light pollution analysis, and AI-powered explanations to provide a personalized celestial observation experience.

Rather than merely displaying celestial objects, the platform focuses on answering practical questions such as:

- What can I see tonight?
- When will the ISS pass overhead?
- Are observation conditions favorable?
- What celestial events are coming next?

Through interactive 3D visualization and predictive simulations, users gain both educational insights and real-world observation guidance.

Core Features

Celestial Dashboard

Upon selecting a location, the platform generates a live dashboard displaying:

- Visible satellites
- International Space Station (ISS) passes

- Planetary positions
- Constellations currently observable
- Moon phase and brightness
- Meteor shower activity
- Upcoming astronomical events

Rather than presenting data as isolated objects, the system organizes information based on visibility and relevance to the user.

Observation Quality Score

One of the platform's key innovations is the Observation Quality Score.

The score evaluates whether a location is suitable for sky observation by combining:

- Cloud cover
- Local weather conditions
- Light pollution
- Moon brightness
- Object elevation angle
- Atmospheric visibility

This feature transforms astronomy data into a practical decision-making tool.

Celestial Timeline

Most astronomy platforms focus only on current positions.

Celestial Lens introduces a timeline simulation system that allows users to explore:

- Current sky conditions
- One hour ahead
- Tomorrow night
- Next week

Users can observe:

- Satellite movement

- Planetary motion
- Moon phase progression
- Future visibility opportunities

This provides both educational value and long-term planning capabilities.

AI Sky Narrator

To make astronomy more accessible, users can request an AI-generated sky summary.

Example:

"Tonight offers excellent visibility conditions. Saturn will become visible in the southwest after 9:13 PM. The ISS is expected to pass overhead in approximately 14 minutes. Jupiter remains visible for most of the evening."

This converts complex scientific information into understandable language.

Impact on Users

Project Zenith makes astronomical information accessible to a broad audience.

For Students:

- Interactive learning of astronomy concepts.
- Better understanding of planetary and satellite movement.

For Amateur Astronomers:

- Observation planning based on real-world visibility conditions.
- Discovery of upcoming celestial events.

For Educators:

- Visual teaching tool for astronomy and space science.

For General Users:

- Simplified explanations of complex celestial phenomena.
- Increased engagement with science and space exploration.

The platform transforms scientific datasets into meaningful insights, encouraging curiosity and public participation in astronomy.

Data Strategy

The platform aggregates data from multiple specialized sources.

Astronomical Data

NASA Horizons:

- Planetary ephemerides
- Astronomical coordinates

AstronomyAPI:

- Planet visibility
- Constellation data
- Moon information

Satellite Data

Open Notify:

- Real-time ISS tracking

N2YO:

- Satellite tracking
- Orbital information

CelesTrak:

- TLE orbital datasets
- Satellite propagation data

Environmental Data

Open-Meteo:

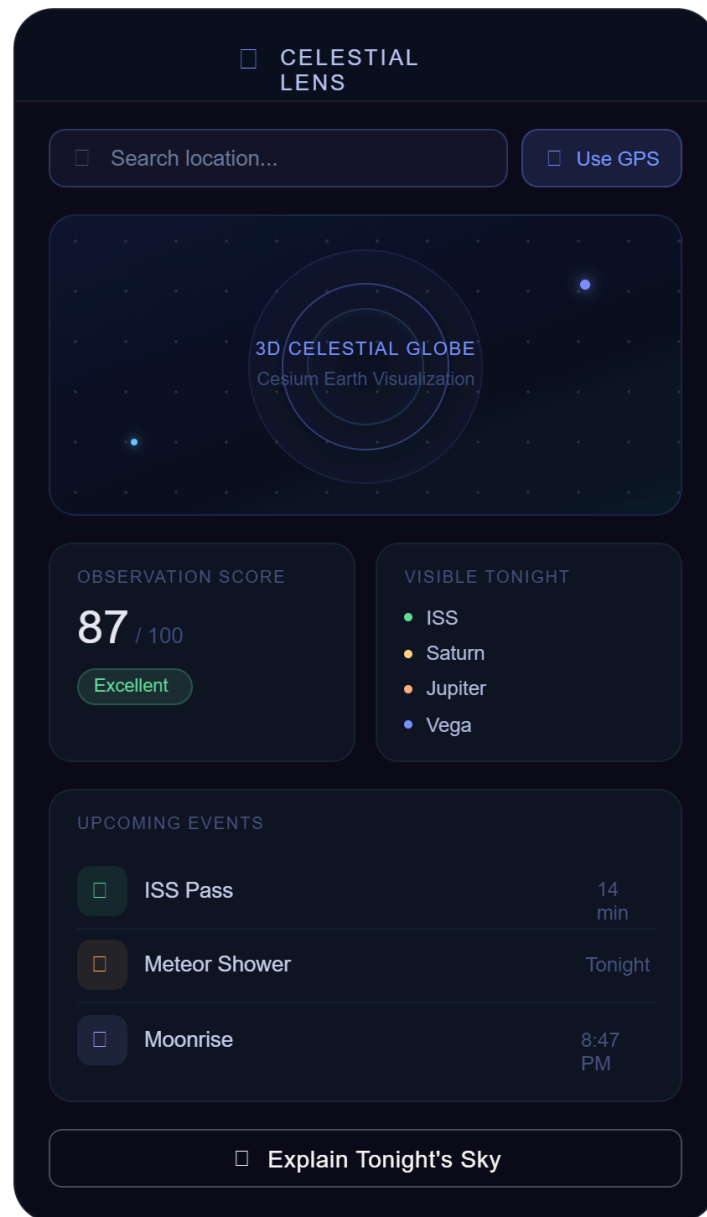
- Cloud cover
- Weather conditions
- Atmospheric visibility

Light Pollution Datasets:

- Night sky quality measurements

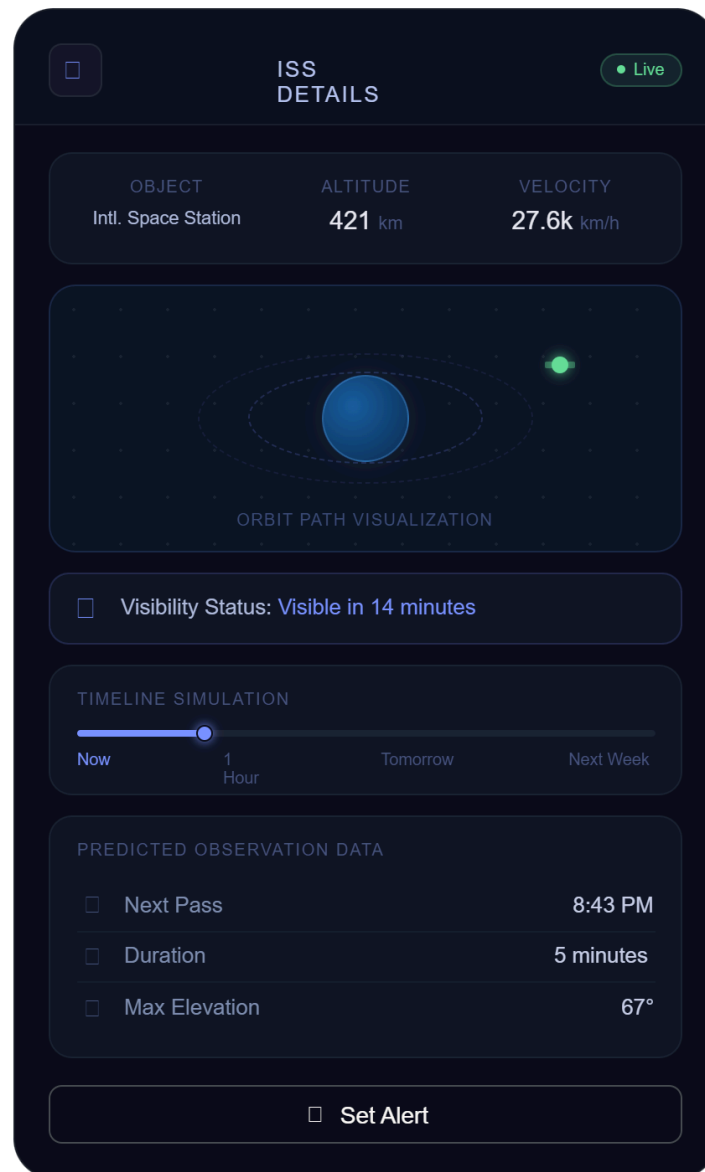
A backend aggregation service will normalize all incoming data into a unified format before delivery to the client.

Wireframe 1 — Celestial Dashboard (Main Screen)



This figure is just a UI mockup, the actual UI Screen might look different.

Wireframe 2 — Object Detail & Timeline View



This figure is just a UI mockup, the actual UI Screen might look different.

Technology Stack

Frontend

- Next.js
- React

- TypeScript
- Tailwind CSS
- shadcn/ui

Visualization

- CesiumJS
- Three.js

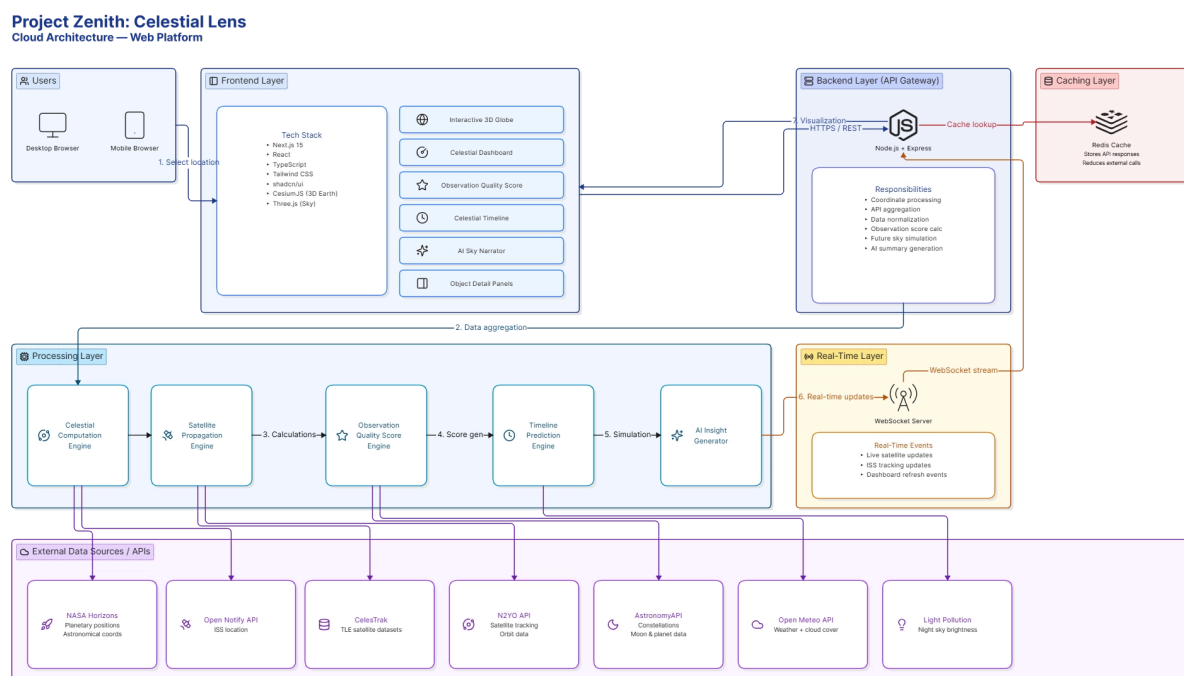
Backend

- Node.js
- Express.js
- Redis Cache

Real-Time Infrastructure

- WebSockets
- Server-side caching
- Scheduled telemetry updates

System Architecture



Approach Towards Development

The development approach follows a modular architecture.

Phase 1 focuses on establishing the data acquisition layer by integrating astronomical, satellite, weather, and light pollution APIs.

Phase 2 focuses on building the data processing layer, including:

- Observation Quality Score Engine
- Satellite Propagation Engine
- Timeline Prediction Engine

Phase 3 focuses on user experience enhancements through:

- Three.js-based 3D visualization
- Interactive dashboards
- AI-generated sky narration

A Redis caching layer and WebSocket infrastructure will be used to support efficient real-time updates and reduce API latency.

Complete Development Roadmap

Day 1

- Project setup
- UI wireframes
- API research and testing

Day 2

- Location services integration
- Data aggregation layer
- Backend API gateway

Day 3

- Satellite tracking implementation
- Planetary visualization
- Observation Quality Score Engine

Day 4

- Timeline simulation engine

- Event prediction system
- Real-time updates using WebSockets

Day 5

- AI Narrator integration
- Dashboard refinement
- Performance optimization

Day 6

- Testing
- Bug fixing
- Deployment preparation

Resources and References

Astronomical Data Sources

- NASA Horizons System
- AstronomyAPI
- Open Notify
- Celestrak
- N2YO

Environmental Data Sources

- Open-Meteo
- Light Pollution Map datasets

Documentation References

- NASA Open Data Documentation
- Open-Meteo API Documentation
- AstronomyAPI Documentation